$2 \sinh x - \tanh^{-1}(\sinh x) + C$

Integral Substitusi Trigonometri

Digunakan dalam integral yang memuat

$\sqrt{a^2-x^2}$, $\sqrt{x^2+a^2}$, $\sqrt{x^2-a^2}$

Caranya dengan substitusi trigonometri yang sesuai.

$\sin^2 x + \cos^2 x = 1^2 \to \cos^2 x = 1^2 - \sin^2 x$

$\to \sin^2 x = 1^2 - \cos^2 x$

↳ Bisa mensubstitusi $\sqrt{a^2-x^2}$

$1^2 + \tan^2 x = \sec^2 x \to \tan^2 x = \sec^2 x - 1$

↳ untuk $\sqrt{x^2+a^2}$ ↳ untuk $\sqrt{x^2-a^2}$

Bentuk Substitusi :

$\sqrt{a^2-x^2} \to x = a\sin\theta \qquad -\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$

$\sqrt{a^2+x^2} \to x = a\tan\theta \qquad -\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$

$\sqrt{x^2-a^2} \to x = a\sec\theta \qquad 0 \le \theta \le \frac{\pi}{2}; x \ge a$

$\pi \le \theta \le \frac{3\pi}{2}; x \le -a$

Contoh

① $\int \frac{1}{x^2\sqrt{4-x^2}}\,dx$

$\int \frac{1}{x^2\sqrt{2^2-x^2}}\,dx$ ($2^2 \to a^2$, $x^2 \to x^2$)

$x = a\sin\theta$

$x = 2\sin\theta \to dx = 2\cos\theta\,d\theta$

$x^2 = 4\sin^2\theta$

$\int \frac{2\cos\theta}{4\sin^2\theta\sqrt{2^2-2^2\sin^2\theta}}\,d\theta$

$\int \frac{2\cos\theta}{4\sin^2\theta\sqrt{2^2\cos^2\theta}}\,d\theta$

$\int \frac{\cancel{2\cos\theta}}{4\sin^2\theta\cdot\cancel{2\cos\theta}}\,d\theta$

$\int \frac{d\theta}{4\sin^2\theta}$

$\frac{1}{4}\int\frac{1}{\sin^2\theta}\,d\theta$

$\frac{1}{4}\int \csc^2\theta\,d\theta = -\frac{1}{4}\cot\theta + C$

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Lalu kembalikan ke nilai x

$x = 2\sin\theta$

$\frac{x}{2} = \sin\theta$

$\theta = \sin^{-1}\left(\frac{x}{2}\right)$

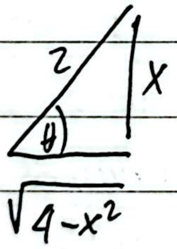
Jadi hasilnya

$-\frac{1}{4}\cot\left(\sin^{-1}\left(\frac{x}{2}\right)\right) + C$

atau dengan segitiga siku ?

$x = 2\sin\theta$

$\sin\theta = \frac{x}{2}$



$\cot\theta = \frac{\sqrt{4-x^2}}{x}$

$-\frac{\sqrt{4-x^2}}{4x} + C$

② $\int \frac{1}{\sqrt{x^2+9}}\,dx$

$\int \frac{1}{\sqrt{\underbrace{x^2}_{x^2} + \underbrace{3^2}_{a^2}}}\,dx$

$x = a\tan\theta$

$x = 3\tan\theta$

$dx = 3\sec^2\theta\,d\theta$

$x^2 = 3^2\tan^2\theta$

$\int \frac{1}{\sqrt{3^2\tan^2\theta + 3^2}}\,3\sec^2\theta\,d\theta$

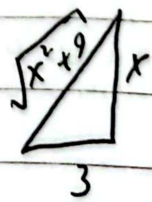
$\int \frac{1}{\sqrt{3^2\sec^2\theta}}\,3\sec^2\theta\,d\theta$

$\int \frac{1}{\cancel{3}\,\cancel{\sec\theta}}\,\cancel{3}\,\overset{\sec\theta}{\cancel{\sec^2\theta}}\,d\theta$

$\int \sec\theta\,d\theta = \ln|\sec\theta + \tan\theta| + C$

$x = 3\tan\theta$

$\frac{x}{3} = \tan\theta$



$\sec\theta = \frac{\sqrt{x^2+9}}{3}$

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$$\ln|\sec\theta+\tan\theta|+C=\ln\left|\frac{\sqrt{x^2+9}}{3}+\frac{x}{3}\right|+C$$
$$=\ln\left|\frac{\sqrt{x^2+9}+x}{3}\right|+C$$

$$\boxed{\begin{aligned}\int\sec x\,dx&=\int\sec x\cdot\frac{\sec x+\tan x}{\sec x+\tan x}\,dx\\&=\int\frac{\sec^2x+\sec x\tan x}{\sec x+\tan x}\,dx\\u&=\sec x+\tan x\\du&=\sec x\tan x+\sec^2x\\&=\int\frac{du}{u}=\ln|u|+C\\&=\ln|\sec x+\tan x|+C\end{aligned}}$$

3. $$\int\frac{\sqrt{1-\ln^2x}}{x}\,dx$$

misal $u=\ln x$
$du=\frac{1}{x}\,dx$

$$\int\sqrt{1-u^2}\,du$$

$$\int\sqrt{1^2-u^2}\,du$$

$u=\sin\theta$
$du=\cos\theta\,d\theta$

$$\int\sqrt{1^2-\sin^2\theta}\cdot\cos\theta\,d\theta$$
$$\int\sqrt{\cos^2\theta}\cdot\cos\theta\,d\theta$$
$$\int\cos\theta\cdot\cos\theta\,d\theta$$
$$\int\cos^2\theta\,d\theta$$
$$=\frac{\theta+\sin\theta\cos\theta}{2}+C$$

$u=\sin\theta$
$\cos\theta=\frac{\sqrt{1-u^2}}{1}$
$\theta=\sin^{-1}u$

$$=\frac{\sin^{-1}u+u\cdot\sqrt{1-u^2}}{2}+C$$
$$=\frac{\sin^{-1}(\ln|x|)+\ln|x|\sqrt{1-\ln^2|x|}}{2}+C$$

Integral Yang Memuat Bentuk Kuadratik

Caranya ubah ax^2+bx+c menjadi bentuk $(x \pm d)^2$ dengan menyempurnakan. Cara menyempurnakannya, ambil $\frac{1}{2}b$ lalu kuadratkan menjadi e mana nanti $c = e + f$, f ini adalah suatu angka sehingga $(x \pm d)^2 + f$

Contoh:

1. $$\int \frac{x}{x^2-4x+8}\,dx$$

$$x^2-4x+8 = x^2-4x+4+4 = (x-2)^2+4$$

$$\int \frac{x}{(x-2)^2+4}\,dx$$

$u = x-2 \rightarrow x = u+2$

$du = dx$

$$\int \frac{u+2}{u^2+4}\,du$$

$$\int \frac{u}{u^2+4}\,du + \int \frac{2}{u^2+4}\,du$$

$$\int \frac{u}{u^2+4}\,du = \frac{1}{2}\int \frac{dw}{w} = \frac{1}{2}\ln|w| + C = \frac{1}{2}\ln|u^2+4| + C = \frac{1}{2}\ln|(x-2)^2+4| + C$$

$w = u^2+4$

$dw = 2u\,du$

$\frac{dw}{2} = u\,du$

$$\int \frac{2}{u^2+4}\,du = \int \frac{2}{(2\tan\theta)^2+4}\,2\sec^2\theta\,d\theta$$

$u = 2\tan\theta$

$du = 2\sec^2\theta\,d\theta$

$\theta = \tan^{-1}\left(\frac{u}{2}\right)$

$$= \int \frac{4\sec^2\theta\,d\theta}{4\tan^2\theta+4}$$

$$= \int \frac{\cancel{4\sec^2\theta}}{\cancel{4\sec^2\theta}}\,d\theta$$

$$= \int 1\,d\theta = \theta + C$$

$$= \tan^{-1}\left(\frac{u}{2}\right) + C$$

$$= \tan^{-1}\left(\frac{x-2}{2}\right) + C$$

$= \frac{1}{2} \ln |(x-2)^2+4| + \tan^{-1}\left(\frac{x-2}{2}\right) + C$

② $\int \frac{1}{\sqrt{5-4x-2x^2}} \, dx$

$5-4x-2x^2 = 5-2(x^2+2x)$

sempurnakan tapi jangan merubah nilai.

$5-2(x^2+2x+1)+2$

$7-2(x+1)^2$

$\int \frac{1}{\sqrt{7-2(x+1)^2}} \, dx$

Misal $u = x+1$

$du = dx$

$\int \frac{1}{\sqrt{7-2u^2}} \, du = \int \frac{1}{\sqrt{2(\frac{7}{2}-u^2)}} \, du$

$= \frac{1}{\sqrt{2}} \int \frac{du}{\sqrt{\frac{7}{2}-u^2}}$

$u = \sqrt{\frac{7}{2}} \sin\theta$

$du = \sqrt{\frac{7}{2}} \cos\theta \, d\theta$

$\frac{1}{\sqrt{2}} \int \frac{\sqrt{\frac{7}{2}} \cos\theta \, d\theta}{\sqrt{\frac{7}{2}-\frac{7}{2}\sin^2\theta}}$

$\frac{1}{\sqrt{2}} \int \frac{\cancel{\sqrt{\frac{7}{2}} \cos\theta} \, d\theta}{\cancel{\sqrt{\frac{7}{2}} \cos\theta}}$

$\frac{1}{\sqrt{2}} \int d\theta = \frac{1}{\sqrt{2}} \theta + C$

$= \frac{1}{\sqrt{2}} \sin^{-1}\left(\sqrt{\frac{2}{7}} u\right) + C$

$= \frac{1}{\sqrt{2}} \sin^{-1}\left(\sqrt{\frac{2}{7}} (x+1)\right) + C$

Integral yg memuat pangkat rasional

Apabila menjumpai variabel / suku dengan nilai

$$x^{1/n}$$

Bisa di substitusi:

$$u = x^{1/m}$$

m adalah kpk dari n

Contoh:

① $$\int \frac{\sqrt{x}}{1+\sqrt[3]{x}}\,dx$$

$$\int \frac{x^{\frac{1}{2}}}{1+x^{\frac{1}{3}}}\,dx$$

kpk $(2,3) = 6$

misal $u = x^{1/6}$

$u^6 = x$

$6u^5\,du = dx$

$$\int \frac{(u^6)^{\frac{1}{2}}}{1+(u^6)^{\frac{1}{3}}} \cdot 6u^5\,du$$

$$6\int \frac{u^8}{1+u^2}$$

$$\begin{array}{r|l} & u^6-u^4+u^2-1 \\ u^2+1 & u^8 \\ & \underline{u^8+u^6} \; - \\ & -u^6 \\ & \underline{-u^6-u^4} \; - \\ & u^4 \\ & \underline{u^4+u^2} \; - \\ & -u^2 \\ & \underline{-u^2-1} \; - \\ & 1 \end{array}$$

$$6\int u^6-u^4+u^2-1+\frac{1}{1+u^2}\,du$$

$$= 6\left[\frac{1}{7}u^7-\frac{1}{5}u^5+\frac{1}{3}u^3-u+\tan^{-1}u\right]+c$$

$$= 6\left[\frac{1}{7}x^{\frac{7}{6}}-\frac{1}{5}x^{\frac{5}{6}}+\frac{1}{3}x^{\frac{3}{6}}-x^{\frac{1}{6}}+\tan^{-1}\left(x^{\frac{1}{6}}\right)\right]+c$$

(6) $\int \sqrt{1+e^x}\, dx$

Misal $u = \sqrt{1+e^x}$

$u^2 = 1+e^x \rightarrow u^2 - 1 = e^x$

$2u\, du = e^x\, dx$

$\frac{2u}{u^2-1}\, du = dx$

$\int u \cdot \frac{2u}{u^2-1}\, du$

$\int \frac{2u^2}{u^2-1}\, du$

$$u^2-1 \overline{\big) 2u^2} \quad 2$$

$$\frac{2u^2-2}{2} -$$

$\int 2 + \frac{2}{u^2-1}\, du$

$2u + \int \frac{2}{(u-1)(u+1)}\, du$

$\frac{2}{(u-1)(u+1)} = \frac{A}{u-1} + \frac{B}{u+1}$

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$\frac{2}{(u-1)(u+1)} = \frac{A(u+1) + B(u-1)}{(u-1)(u+1)}$

$2 = A(u+1) + B(u-1)$

$u = 1 \rightarrow$

$2 = A(2) + B(0)$

$A = 1$

$u = -1 \rightarrow$

$2 = A(0) + B(-1-1)$

$2 = -2B$

$B = -1$

$2u + \int \frac{1}{u-1} - \frac{1}{u+1}\, du$

$= 2u + \ln|u-1| - \ln|u+1| + C$

$= 2\sqrt{1+e^x} + \ln\left|-1+\sqrt{1+e^x}\right| - \ln\left|1+\sqrt{1+e^x}\right| + C$